Parallel Global Convolution-Enhanced Spatial Chunk Transformer Network for Few-Shot Hyperspectral Image Classification

Jun Pang, Lina Yang, *Member, IEEE*, Xinzhang Wu, *Senior Member, IEEE*, Wanxing Zhu, and Lianlian Zhao

***Abstract*—Hyperspectral image (HSI) classification under few-shot scenarios remains a challenging task due to the high dimensionality of HSI data and the scarcity of labeled samples. Traditional convolutional neural networks (CNNs) struggle to capture long-range spectral–spatial dependencies due to limited local receptive fields, while pure Transformers suffer from $O((HW)^2)$ high computational complexity and weak local feature extraction capability, making them poorly adaptive to few-shot HSI classification. To address these issues, we propose a parallel global convolution-enhanced spatial chunk Transformer network (PGC-SCTN), which realizes progressive spectral–spatial feature optimization via three core modules. Specifically, a deep convolutional embedding (DCE) module mines high-dimensional local fine-grained spectral–spatial features of HSI. A parallel global feature enhancement extraction (PGFE) module then adopts horizontal–vertical separated convolution to efficiently capture long-range spectral–spatial dependencies, compensating for the global modeling deficiency of CNNs. Furthermore, a spatial chunk Transformer (SCT) module decomposes 2-D self-attention into 1-D chunk computation along the height and width directions, reducing the computational complexity of global dependency modeling from $O((HW)^2)$ to $O(HW*(H + W))$ and balancing global semantic refinement with computational efficiency. Experiments on four public hyperspectral datasets show that the proposed network outperforms traditional methods and state-of-the-art (SOTA) algorithms in classification performance, demonstrating superior robustness and generalization.**

***Index Terms*—Convolutional neural network (CNN), hyperspectral image (HSI) classification, parallel global convolution (PGC), spatial chunk Transformer (SCT).**

I. Introduction

HYPERSPECTRAL Imaging, an advanced remote sensing technique, captures high-resolution "spectrum-spatial integration" information beyond human vision via hundreds of contiguous narrow bands, accurately recording ground objects' spectral curves and spatial reflectance [1], [2], [3]. It thus acts as a core technical pillar in environmental monitoring, urban planning, mineral exploration, and ecological assessment [4], [5], [6], [7]. As hyperspectral image (HSI) data processing's key task, classification assigns land cover categories to each pixel for refined recognition [8], [9]. However, HSI's high dimensionality and spectral redundancy pose critical challenges: while rich spectral data benefits classification, redundant dimensions trigger the "curse of dimensionality," raising computational costs and degrading traditional models' accuracy and generalization [10], [11]. Balancing high-dimensional data processing and spectral mining is therefore a central focus of current remote sensing research [12], [13].

To address HSI data redundancy from high-resolution imaging, statistical key-band extraction and dimensionality reduction are core solutions [14], divided into two spectral transformation branches: nonlinear and linear. Nonlinear methods excel at capturing HSI's inherent nonlinear features. A representative method within this category is kernel discriminant analysis (KDA) [15], which effectively models the complex spectral–spatial correlations of ground objects. Computationally efficient linear methods fall into two categories: discriminant transformations for enhanced class separability, e.g., linear discriminant analysis (LDA) [16], and orthogonal transformations for redundancy elimination via orthogonalization, notably principal component analysis (PCA) [17] and singular value decomposition (SVD) [18]. PCA is the most widely used classic technique in this category.

HSI classification has undergone a significant shift from traditional machine learning to deep learning [19], [20], [21], and the latter has driven breakthroughs in HSI classification through its automatic feature learning capability. With the application of methods such as stacked autoencoders (SAEs) [22], recurrent neural networks (RNNs) [23], graph convolutional networks (GCNs) [24], and convolutional neural networks (CNNs) [25] in HSI classification. These deep learning approaches have substantially improved classification accuracy compared to traditional methods.

CNNs dominate HSI classification due to their efficient local feature extraction capability [26], [27]. With the development of deep learning, optimizing "feature representation

Received 7 January 2026; revised 1 April 2026 and 7 May 2026; accepted 22 May 2026. Date of publication 26 May 2026; date of current version 2 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62371144 and Grant 62461004 and in part by Guangxi Science and Technology Major Program under Grant AA23073019. *(Corresponding authors: Lina Yang; Xinzhang Wu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3696991

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ability" has become a research focus, and fusion strategies involving multiscale training, residual structures, and attention mechanisms have garnered significant attention. Following these directions, Xie et al. [28] designed a multiscale dense convolutional network (MS-DenseNet) framework. The framework constructs training and testing samples by extracting multiscale patches, fully explores the complementarity and correlation of information between network layers through dense blocks, and further incorporates multiple dense blocks to achieve deep fusion of multiscale information. In contrast, Zhang et al. [29] proposed a novel optimization direction for the spectral dimensional features of HSI. They developed a CNN-based efficient spectral partitioning residual network (SPRN), which first splits the input spectral bands into nonoverlapping continuous subbands and then employs cascaded parallel improved residual blocks for comprehensive spectral–spatial feature extraction.

The Transformer architecture, leveraging its unique long-range sequence processing capability and multihead self-attention (MHSA) mechanism, offers a new direction to overcome the limitations of traditional methods [30]. Unlike CNNs, which excel at extracting local features but struggle to model spatial context information. Transformers effectively capture global contextual correlations—making them a research hotspot in the recent HSI classification field [31]. In early applications, Hong et al. [32] proposed Spectralformer, a backbone network for mining spectral feature sequences from a sequential perspective, significantly improving classification performance. Zhou et al. [33] further combined RNNs with Transformers to design a segmentally recursive RT model, addressing the challenges of sequence data processing. Although the standard Transformer architecture excels at modeling long-range dependencies, it inherently lacks an explicit local bias, causing the self-attention mechanism to overlook fine-grained local spectral–spatial relationships at the pixel level. To address this issue, Su et al. [34] proposed the self-supervised relation-aware vision Transformer (SRViT), which incorporates relation modeling to enhance local feature learning for hyperspectral unmixing.

In recent years, hybrid CNN–Transformer methods have become the mainstream paradigm for HSI classification and have achieved state-of-the-art (SOTA) performance. The method leverages the advantages of CNNs for local spectral–spatial feature extraction and Transformers for global dependency modeling, enabling complementary local-to-global feature learning. Sun et al. [35] proposed a spectral–spatial feature tokenization Transformer (SSFTT), which jointly captures spectral–spatial features and high-level semantics using CNN-based local feature encoding and Transformer-based global semantic modeling. To uncover local and hierarchical spatial–spectral relationships, Xu et al. [36] developed a spatial–spectral 1-D Swin Transformer (SS1DSwin), which combines CNNs with window-based self-attention to balance local feature refinement and global dependency modeling. Notably, local structure modeling in Transformer-based HSI models is not restricted to CNNs. Recent studies have explored various advanced strategies for this purpose, among which graph-based Transformer models represent a prominent branch. Jiang et al. [37] presented a graph generative structure-aware Transformer (GraphGST), an architecture that achieves fine-grained local structure modeling of HSI data via graph representation and then fuses it with the global attention of Transformers for end-to-end HSI classification.

The aforementioned deep learning-based end-to-end adaptive learning algorithms have significantly improved hyperspectral classification accuracy, yet high-precision classification in few-shot scenarios still poses substantial challenges. To address this, Xu et al. [38] proposed RSEN, which achieves self-ensemble via mutual learning between base and ensemble networks. However, existing few-shot methods lack feature discrimination and overfitting resistance, limiting their effectiveness. Liu et al. [39] thus proposed SSEFN, with strong feature extraction and few-shot overfitting robustness. Despite their excellent performance in few-shot HSI classification, current CNN-based and CNN–Transformer hybrid methods still exhibit limitations in this task.

1) Traditional CNNs struggle to capture long-range spectral–spatial dependencies due to their limited local receptive fields, while pure Transformers—though adept at modeling global correlations—underperform in extracting local fine-grained features. Neither approach effectively balances the efficient learning of both core feature types.
2) The global attention mechanism of traditional Transformers has a high complexity of $O((HW)^2)$, making it hard to balance computational efficiency and model performance in few-shot scenarios due to poor adaptability.
3) Traditional handcrafted methods have performance bottlenecks, while some existing deep learning algorithms tend to suffer from low feature discrimination and high overfitting risks in few-shot scenarios, limiting classification accuracy and generalization.

To address the aforementioned issues, this article proposes the parallel global convolution-enhanced spatial chunk Transformer network (PGC-SCTN), a novel few-shot HSI classification framework. Specifically, targeting CNNs' strength in local feature extraction but weakness in global modeling, we design a deep convolutional embedding (DCE) module to accurately mine local fine-grained spectral–spatial features from high-dimensional HSI data via deep convolution. A parallel global feature enhancement extraction (PGFE) module then adopts a horizontal–vertical separable structure to efficiently capture long-range spectral–spatial correlations and break traditional CNNs' receptive field limitation. Furthermore, aiming at the pure Transformers' $O((HW)^2)$ high complexity incompatibility with few-shot scenarios, a spatial chunk Transformer (SCT) module decomposes 2-D global attention into 1-D chunk computations along height and width, reducing computational complexity to O(HW*(H + W)) while refining global semantics and improving efficiency for scarce labeled samples. Experiments on four public HSI datasets show PGC-SCTN significantly outperforms traditional and SOTA deep learning methods in few-shot classification, with

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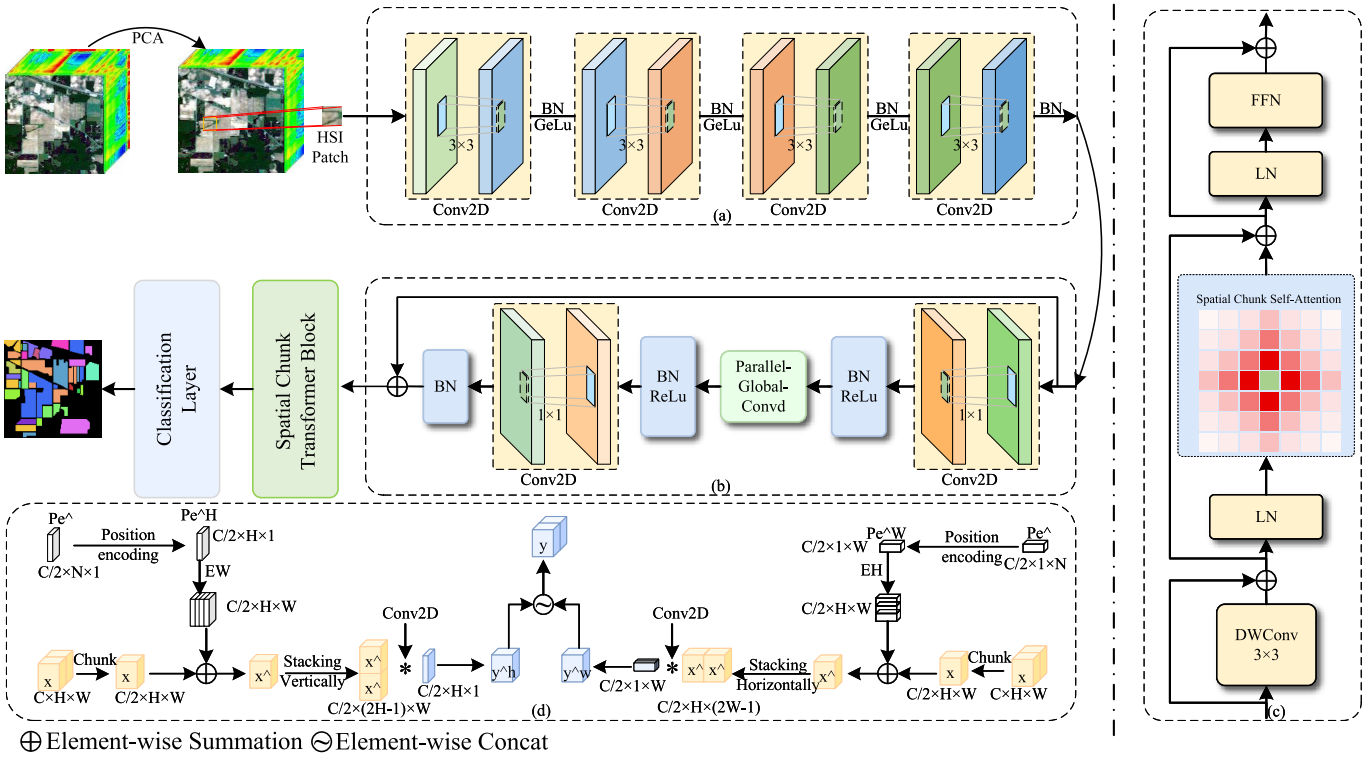


Fig. 1. Overview of the proposed PGC-SCTN. (a) Deep convolution embedding module. (b) PGFE module. (c) SCT. (d) PGC module.

superior robustness and generalization. Our contributions are summarized as follows.

1) We propose a deep convolution embedding module. Specifically, it accurately mines local fine-grained spectral–spatial features from high-dimensional HSI data through deep convolution operations.
2) We propose a PGFE module. In the global dimension, it adopts a horizontal–vertical separable convolution structure to efficiently capture long-range spectral–spatial dependencies of HSI, making up for the global modeling deficiency of traditional CNNs.
3) In contrast to conventional 2-D self-attention with quadratic complexity, the proposed SCT decomposes global attention into 1-D modeling along the height chunks and width chunks of the spatial axis, reducing computational complexity while preserving global semantic consistency.

The remainder of the article is organized as follows. Section II elaborates on the proposed algorithms. Section III outlines the comprehensive comparative experiments and ablation experiments. Section IV concludes the work and discusses future directions.

II. Proposed Method

Fig. 1 illustrates the overall framework of the proposed PGC-SCTN. Given an input HSI $\mathbf{I} \in \mathbb{R}^{B\times H\times W}$, where B, H, and W denote the number of spectral bands, image height, and image width, respectively. A PCA preprocessing step is first applied to alleviate spectral redundancy. After dimensionality reduction, the input is transformed into $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, where C denotes the reduced spectral dimensionality.

Assume that the HSI contains N labeled pixels $\mathcal{X}_N = x_1, x_2, \ldots, x_N$ with corresponding one-hot labels $\mathcal{Y}_N = y_1, y_2, \ldots, y_N$. To jointly exploit spectral and spatial information, a fixed-size spatial neighborhood of $t \times t$ pixels centered at each labeled pixel is extracted, forming an input sample set $\mathcal{Z}_N = z_1, z_2, \ldots, z_N \in \mathbb{R}^{C\times t\times t}$. These spectral–spatial patches are subsequently fed into the proposed network for feature learning and classification.

A. Deep Convolution Embedding Module

In few-shot HSI classification, insufficient labeled samples often lead to inadequate mining of local spectral–spatial details. To address this issue, a deep convolution embedding (DCE) module is designed to effectively extract fine-grained local features from high-dimensional input patches. The DCE module adopts a cascaded convolutional architecture combined with batch normalization (BN) and Gaussian error linear unit (GeLU) activation to enhance feature discriminability while mitigating overfitting.

Fig. 1(a) illustrates the overall framework of the proposed DCE. The input block Z_i first undergoes a 2-D convolution operation to capture local spatial correlations, followed by BN and GeLU activation to introduce nonlinearity. A single convolutional operation can be formulated as

$$\mathrm{Y} = \mathrm{GeLU}\left(\mathrm{BN}\left(\mathrm{Conv2d}_{3\times 3}\left(Z_i; W, b\right)\right)\right) \tag{1}$$

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where $\mathrm{Conv2D}_{3\times3}(\cdot)$ denotes the 2-D convolution with a 3×3 kernel, BN represents batch normalization, and GeLU is the Gaussian error linear unit activation function.

To progressively extract hierarchical local representations, multiple convolutional layers are stacked, enabling the network to capture increasingly abstract spectral–spatial patterns. The overall DCE module can be expressed as

$$\mathrm{X}_{\mathrm{local}} = f_4 \circ g_3 \circ f_3 \circ g_2 \circ f_2 \circ g_1 \circ \tag{2}$$

where $f_i(\cdot)$ denotes a convolution operation and $g_i(\cdot)$ represents the combined BN and GeLU activation. The output $\mathbf{X}_{\mathrm{local}}$ corresponds to the extracted local fine-grained spectral–spatial features. By focusing on low-level details such as edge structures and local spectral variations, the DCE module provides a robust foundation for subsequent global feature modeling.

B. PGFE Module

While CNNs are effective in modeling local patterns, their limited receptive fields hinder the capture of long-range spectral–spatial dependencies. To overcome this limitation, a PGFE module is introduced, which integrates a parallel global convolution (PGC) mechanism with residual feature fusion.

In this section, we first introduce the PGC module, as illustrated in Fig. 1(d). The proposed PGC module employs a horizontal–vertical separable structure to efficiently model global dependencies. Given the input feature map $\mathbf{X}'_{\mathrm{local}} \in \mathbb{R}^{B_s\times C\times t\times t}$, the channel dimension is evenly split into two branches corresponding to horizontal (X^{h}) and vertical (X^{w}) modeling

$$\begin{aligned}\mathrm{X}^{\mathrm{h}}, \mathrm{X}^{\mathrm{w}} &= \mathrm{Chunk}\left(\mathrm{X}'_{\mathrm{local}};\ \mathrm{dim} = C,\ k = 2\right)\\ \mathrm{X}^{\mathrm{h}} &\in \mathbb{R}^{B_s\times\frac{C}{2}\times t\times t},\quad \mathrm{X}^{\mathrm{w}} \in \mathbb{R}^{B_s\times\frac{C}{2}\times t\times t}\end{aligned} \tag{3}$$

where B_s denotes the batch size, C represents the number of input channels, and Chunk denotes a splitting operator that equally partitions the tensor into k parts along the dim dimension.

To enhance spatial awareness, learnable positional encodings are injected into each branch, allowing the model to adaptively capture spatial layout variations among different ground objects

$$\begin{cases}\mathrm{X}^{\mathrm{h}}_{\mathrm{pe}} = \mathrm{X}^{\mathrm{h}} + \mathcal{P}^{\mathrm{h}} \otimes \mathrm{l}_{1\times t}, & \mathcal{P}^{\mathrm{h}} \in \mathbb{R}^{1\times\frac{C}{2}\times t\times 1}\\ \mathrm{X}^{\mathrm{w}}_{\mathrm{pe}} = \mathrm{X}^{\mathrm{w}} + \mathrm{l}_{t\times1} \otimes \mathcal{P}^{\mathrm{w}}, & \mathcal{P}^{\mathrm{w}} \in \mathbb{R}^{1\times\frac{C}{2}\times 1\times t}\end{cases} \tag{4}$$

where $\mathcal{P}^{\mathrm{h}}/\mathcal{P}^{\mathrm{w}}$ denote the horizontal/vertical learnable positional encoding, $\otimes$ denotes the broadcast outer product (enabling dimension matching between positional encoding and feature maps), and $\mathrm{l}_{a\times b}$ represents an a $\times$ b-dimensional all-ones matrix.

To enhance the spatial dependency modeling capability for HSI, a direction-specific concatenation strategy is adopted to expand the dimension of feature maps. Specifically, the feature map is expanded to size $2t-1$ along the height or width direction by concatenating the original feature with its first $t-1$ rows or columns. This expansion ensures that the subsequent direction-specific convolution kernel with size $t\times1$ or $1\times t$ can fully cover the entire spatial range of the original $t\times t$ patch, thereby achieving complete and consistent global spatial dependency modeling along the horizontal or vertical direction. The process can be expressed as

$$\begin{aligned}&\begin{cases}\mathrm{X}^{\mathrm{h}}_{\mathrm{cat}} = \mathrm{Concat}\left(\mathrm{X}^{\mathrm{h}}_{\mathrm{pe}}, \mathrm{X}^{\mathrm{h}}_{\mathrm{pe}}[:,:,1:t,:];\mathrm{dim} = \mathrm{h}\right)\\ \mathrm{X}^{\mathrm{w}}_{\mathrm{cat}} = \mathrm{Concat}\left(\mathrm{X}^{\mathrm{w}}_{\mathrm{pe}}, \mathrm{X}^{\mathrm{w}}_{\mathrm{pe}}[:,:,:,1:t];\mathrm{dim} = \mathrm{w}\right)\end{cases}\\ &\mathrm{X}^{\mathrm{h}}_{\mathrm{cat}} \in \mathbb{R}^{B_s\times\frac{C}{2}\times(2t-1)\times t},\quad \mathrm{X}^{\mathrm{w}}_{\mathrm{cat}} \in \mathbb{R}^{B_s\times\frac{C}{2}\times t\times(2t-1)}\end{aligned} \tag{5}$$

where h/w denote the height/width dimensions of the feature map, respectively; $\mathrm{X}[:,:,1:t,:]$ represents extracting rows 1 to t − 1 (excluding the last row) along the height dimension of the feature map; $\mathrm{Concat}(\cdot;dim)$ is a tensor concatenation operator along the specified dimension.

Focusing on the core of horizontal/vertical collaborative modeling for HSI, 2-D convolution is applied to achieve direction-specific feature extraction, which efficiently models the horizontal/vertical global dependencies and provides reliable spatial feature support for ground object category determination. The process can be expressed as

$$\begin{aligned}&\begin{cases}\mathrm{Y}^{\mathrm{h}} = \mathrm{Conv2d}\left(\mathrm{X}^{\mathrm{h}}_{\mathrm{cat}}; \mathcal{W}^{\mathrm{h}}, b^{\mathrm{h}}, K^{\mathrm{h}}\right)\\ \mathrm{Y}^{\mathrm{w}} = \mathrm{Conv2d}\left(\mathrm{X}^{\mathrm{w}}_{\mathrm{cat}}; \mathcal{W}^{\mathrm{w}}, b^{\mathrm{w}}, K^{\mathrm{w}}\right)\end{cases}\\ &\mathrm{Y}^{\mathrm{h}} \in \mathbb{R}^{B_s\times\frac{C_{\mathrm{out}}}{2}\times t\times t}, \mathrm{Y}^{\mathrm{w}} \in \mathbb{R}^{B_s\times\frac{C_{\mathrm{out}}}{2}\times t\times t}\end{aligned} \tag{6}$$

where $\mathcal{W}^{\mathrm{h}}/\mathcal{W}^{\mathrm{w}}$ and $b^{\mathrm{h}}/b^{\mathrm{w}}$ denote the weights and biases of the horizontal/vertical convolutions, respectively; $K^{\mathrm{h}} = (t,1)$ and $K^{\mathrm{w}} = (1,t)$ represent the sizes of the direction-specific convolution kernels; C_{out} denotes the total number of output channels.

Finally, the convolution outputs of the horizontal/vertical branches are concatenated along the channel dimension to aggregate the global features from both directions

$$\mathrm{X}_{\mathrm{PGC}} = \mathrm{Concat}\left(\mathrm{Y}^{\mathrm{h}}, \mathrm{Y}^{\mathrm{w}};\mathrm{dim} = C\right) \in \mathbb{R}^{B_s\times C_{\mathrm{out}}\times t\times t} \tag{7}$$

where $\mathrm{Y}^{\mathrm{h}}/\mathrm{Y}^{\mathrm{w}}$ are the intermediate outputs of the horizontal/vertical branches; through concatenation, the bidirectional feature complementarity is achieved to form the final output $\mathrm{X}_{\mathrm{PGC}}$.

Algorithm 1 Procedure of the PGC Module

Input: The input feature map $\mathbf{X}'_{\mathrm{local}} \in \mathbb{R}^{B_s\times C\times t\times t}$;
Output: PGC feature map $\mathbf{X}_{\mathrm{PGC}} \in \mathbb{R}^{B_s\times C_{\mathrm{out}}\times t\times t}$.
Forward:
$\mathbf{X}^{\mathrm{h}}, \mathbf{X}^{\mathrm{w}} = \mathrm{Chunk}(\mathbf{X}'_{\mathrm{local}})$
$\mathbf{X}^{\mathrm{h}}_{\mathrm{pe}} = \mathbf{X}^{\mathrm{h}} + \mathrm{Expand}\left(\mathcal{P}^{\mathrm{h}}, \left(1, \frac{C}{2}, t, t\right)\right)$
$\mathbf{X}^{\mathrm{w}}_{\mathrm{pe}} = \mathbf{X}^{\mathrm{w}} + \mathrm{Expand}\left(\mathcal{P}^{\mathrm{w}}, \left(1, \frac{C}{2}, t, t\right)\right)$
if type == horizontal operation **then**
  $\mathbf{X}^{\mathrm{h}}_{\mathrm{cat}} = \mathrm{Concat}(\mathbf{X}^{\mathrm{h}}_{\mathrm{pe}}, \mathbf{X}^{\mathrm{h}}_{\mathrm{pe}}[:,:,1:t,:], \mathrm{dim} = 2)$
  $\mathbf{Y}^{\mathrm{h}} = \mathrm{conv2d}(\mathbf{X}^{\mathrm{h}}_{\mathrm{cat}})$
else
  $\mathbf{X}^{\mathrm{w}}_{\mathrm{cat}} = \mathrm{Concat}(\mathbf{X}^{\mathrm{w}}_{\mathrm{pe}}, \mathbf{X}^{\mathrm{w}}_{\mathrm{pe}}[:,:,:,1:t], \mathrm{dim} = 3)$
  $\mathbf{Y}^{\mathrm{w}} = \mathrm{conv2d}(\mathbf{X}^{\mathrm{w}}_{\mathrm{cat}})$
end if
$\mathbf{X}_{\mathrm{PGC}} = \mathrm{Concat}(\mathbf{Y}^{\mathrm{h}}, \mathbf{Y}^{\mathrm{w}}; dim = 1)$
Global spectral–spatial features of $\mathbf{X}_{\mathrm{PGC}}$

By adopting a horizontal–vertical separable convolution structure, the module effectively bridges the global

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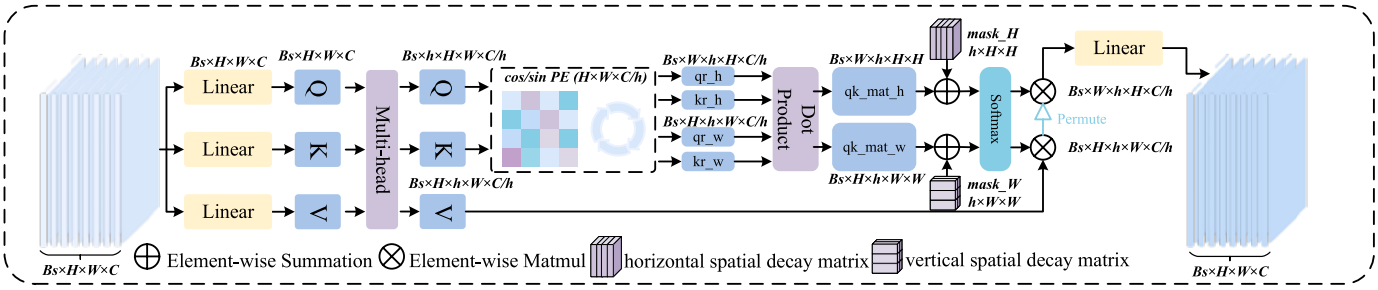


Fig. 2. Architecture of the SCSA. The input HSI feature matrix is linearly projected to produce the Q/K/V matrices, which are then reshaped for multihead attention. Rotary positional encoding is applied to the queries and keys. We first conduct vertical (width) attention with a vertical exponential decay mask (mask_W), followed by horizontal (height) attention with a horizontal exponential decay mask (mask_H). The resulting features are fed into a linear layer to yield the final extracted features.

dependency gap in local features and enables complementary learning of local details and global correlations (see Algorithm 1).

Fig. 1(b) illustrates the overall framework of the proposed PGFE. To avoid the loss of local details that may be caused by PGC, we design a progressive processing pipeline of "shallow extraction—global modeling—information integration—residual fusion." Specifically, the input features undergo shallow feature extraction through a 1×1 2-D convolution layer. Subsequently, they are fed into the PGC module to efficiently capture the global spectral–spatial information of HSI. Furthermore, the global information is integrated and optimized via another 1×1 2-D convolution layer. Finally, a residual connection is performed with the original input feature map, achieving balanced modeling of local details and global correlations. The process can be expressed as

$$\mathrm{X}'_{\text{local}} = \delta\left(\mathrm{BN}\left(\mathrm{Conv2d}_{1\times1}\left(\mathrm{X}_{\text{local}}\right)\right)\right) \tag{8}$$

$$\mathrm{X}_{\text{PGC}} = \delta\left(\mathrm{BN}\left(\mathrm{ParGC}\left(\mathrm{X}'_{\text{local}}\right)\right)\right) \tag{9}$$

$$\mathrm{X}_{\text{global}} = \mathrm{X}_{\text{local}} + \mathrm{BN}\left(\mathrm{Conv2d}_{1\times1}\left(\mathrm{X}_{\text{PGC}}\right)\right) \tag{10}$$

where δ refers to the nonlinear activation function ReLU, and ParGC denotes the parallel global convolution operation. $\mathrm{X}_{\text{global}}$ is the global feature extracted by the PGFE module and serves directly as the input to the SCT module. This feature map first undergoes a depthwise separable convolution (DWConv) residual enhancement operation to refine local spatial details before fed into the spatial chunk self-attention (SCSA) unit of the SCT module.

C. Spatial Chunk Transformer

Traditional Transformer architectures suffer from quadratic computational complexity with respect to spatial resolution, which limits their applicability in few-shot HSI scenarios. To address this issue, we design an SCT module, as illustrated in Fig. 1(c), which decomposes 2-D global attention into 1-D computations over height chunks and width chunks to balance global semantic refinement and computational efficiency.

In this section, we first introduce the SCSA unit [Fig. 2 shows the overall architecture of the SCSA.]. To enhance spatial modeling, inspired by retention networks [40], we incorporate rotary position encoding (RoPE) and a distance-aware decay mask based on Manhattan distance into the attention computation. RoPE enables relative positional awareness, while the decay mask imposes a spatial prior. Given the query and key vectors at positions i and j, the rotary transformation can be expressed as

$$\tilde{\mathrm{Q}}_i = \mathrm{R}\,(i)\,\mathrm{Q}_i, \quad \tilde{\mathrm{K}}_j = \mathrm{R}\,(j)\,\mathrm{K}_j \tag{11}$$

where $\mathrm{R}(\cdot)$ denotes the rotation matrix constructed using sine and cosine functions.

We introduce a distance-aware decay mask to impose an explicit spatial prior on the attention mechanism. It can be expressed as

$$\begin{aligned} &\mathrm{Q} = \left(XW_Q\right) \odot \Theta, \mathrm{K} = \left(XW_K\right) \odot \bar{\Theta}, \mathrm{V} = XW_V \\ &\Theta_n = e^{in\theta}, \; D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases} \end{aligned} \tag{12}$$

where $\bar{\Theta}$ is the complex conjugate of Θ, $D \in \mathbb{R}^{|x|\times|x|}$ denotes the distance-aware decay masks, which capture the relative distances within 1-D sequences.

In addition, we propose a direct and efficient method that extends the unidirectional retention to a bidirectional form, adapting to the processing needs of high-dimensional spectral–spatial data, expressed as follows:

$$D^{Bi}_{nm} = \gamma^{|n-m|} \tag{13}$$

$$\mathrm{BiRetention}\,(X) = \left(\mathrm{softmax}\left(\frac{\tilde{\mathrm{Q}}\tilde{\mathrm{K}}^T}{\sqrt{d_K}} \odot D^{Bi}_{nm}\right)\right)\mathrm{V} \tag{14}$$

where BiRetention signifies bidirectional modeling, and d_K is the dimension of K.

Though the bidirectional BiRetention solves the 1-D sequence modeling problem, it is still insufficient for 2-D grid-structured HSI data, where each pixel token is positioned with 2-D spatial coordinates (x_n, y_n) (the nth token). To adapt to 's 2-D spatial characteristics, we redefine the spatial decay matrix D to characterize the Manhattan distance between any two pixel tokens in the 2-D plane, which quantifies the spatial correlation prior between ground object pixels in HSI. The 2-D spatial decay matrix and corresponding 2-D attention are expressed as follows:

$$\begin{aligned} \mathrm{Attention}\,(X) &= \left(\mathrm{softmax}\left(\frac{\tilde{\mathrm{Q}}\tilde{\mathrm{K}}^T}{\sqrt{d_K}} \odot D^{2d}_{nm}\right)\right)\mathrm{V} \\ D^{2d}_{nm} &= \gamma^{|x_n - x_m| + |y_n - y_m|}. \end{aligned} \tag{15}$$

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Directly computing the above 2-D attention for high-resolution HSI feature maps leads to quadratic computational complexity $O((HW)^2)$, which is not suitable for few-shot HSI classification with scarce labeled samples and limited computational resources. To address this issue, we introduce a dual 1-D decomposition strategy for both the 2-D self-attention computation and the 2-D spatial decay matrix: the global 2-D attention is decomposed into independent horizontal (height) and vertical (width) 1-D chunkwise attention computations, and the 2-D spatial decay matrix is split into horizontal and vertical 1-D decay matrices that characterize the spatial distance between tokens in a single direction. Specifically, we calculate the attention scores in the horizontal and vertical directions of the HSI separately. Subsequently, we apply 1-D bidirectional decay matrices to these attention weights, where the 1-D decay matrices represent the horizontal and vertical distances between tokens ($D^H_{nm} = \gamma^{|y_n - y_m|}, D^W_{nm} = \gamma^{|x_n - x_m|}$)

$$\mathrm{Attn}_H = \mathrm{Softmax}\left(\frac{\tilde{\mathrm{Q}}_H \tilde{\mathrm{K}}_H^T}{\sqrt{d_K}}\right) \odot D^H_{nm} \tag{16}$$

$$\mathrm{Attn}_W = \mathrm{Softmax}\left(\frac{\tilde{\mathrm{Q}}_W \tilde{\mathrm{K}}_W^T}{\sqrt{d_K}}\right) \odot D^W_{nm} \tag{17}$$

$$\mathrm{SCSA}(X) = \mathrm{Attn}_H(\mathrm{Attn}_W \mathrm{V})^T \tag{18}$$

where $\tilde{\mathrm{Q}}_H$, $\tilde{\mathrm{K}}_H$, and $\tilde{\mathrm{Q}}_W$, $\tilde{\mathrm{K}}_W$ are the query and key matrices projected for the horizontal and vertical directions, respectively. The serial computation of horizontal and vertical 1-D attention realizes the efficient modeling of global 2-D spatial correlations for HSI and reduces the computational complexity from $O((HW)^2)$ to O(HW*(H + W)).

To enhance local features, capture local spatial characteristics, and compensate for the shortcomings of global attention, we perform residual DWConv feature extraction before $\mathrm{X}_{\mathrm{global}}$ passes through the SCSA unit

$$\mathrm{X}'_{\mathrm{global}} = \mathrm{X}_{\mathrm{global}} + \mathrm{DWConv}\left(\mathrm{X}_{\mathrm{global}}\right). \tag{19}$$

Compared with traditional Transformer architectures, the proposed SCT mechanism—by decomposing self-attention computation along the horizontal and vertical directions of HSI—achieves dual advantages: it balances the optimization of global spectral–spatial semantics and reduces computational complexity from $O((HW)^2)$ to O(HW*(H + W)). Especially in few-shot scenarios with scarce labeled samples, it better balances computational efficiency and model performance, addressing the poor adaptability of traditional architectures in high-dimensional HSI processing.

D. Final Classification Module

The feature tensor output by the PGC-SCTN backbone is first aggregated into a 1-D feature vector of dimension B × C via global average pooling (GAP) over the spatial dimensions (t × t), which compresses spatial information while retaining the global spectral–spatial feature of each pixel patch. Subsequently, the 1-D feature vector is fed into the classification module, which consists of a layer normalization layer followed by a fully connected linear layer—a design that stabilizes feature distribution before classification and avoids internal covariate shift while directly mapping the feature vector to the number of target categories for HSI classification. The mathematical expression of the classification module is

$$\hat{y} = \mathrm{Linear}\left(\mathrm{LayerNorm}\left(\mathrm{GAP}\left(\mathrm{X}_{\mathrm{feat}}\right)\right)\right) \tag{20}$$

where $\mathrm{X}_{\mathrm{feat}}$ denotes the feature tensor output by the PGC-SCTN backbone, GAP(·) represents global average pooling over spatial dimensions, LayerNorm(·) is layer normalization with the normalized dimension set to C, Linear(·) is a fully connected layer with input dimension C and output dimension equal to the number of ground object categories in the target HSI dataset, and $\hat{y}$ is the predicted category probability vector for each pixel patch.

III. Experimental Results and Analysis

In the present experiments, we utilize four distinct datasets to comprehensively evaluate the performance of the proposed HSI classification model: Indian Pines (IP), Salinas Scene (SA), Houston 2013 (HU), and Fanglu Tea Farm (FLT). Table I shows the surface coverage details and experimental settings of the four datasets.

A. Data Description

1) Indian Pine: The IP dataset, captured by the airborne visual infrared imaging spectrometer (AVIRIS) in 1992 for HSI classification, covers a 145 × 145 area of Indiana's Indian Pine trees (20 m spatial resolution). It retains 200 spectral bands (from the original 220 after removing water-absorbing bands), includes 16 vegetation classes, and has 10 249 labeled pixels in total.

2) Salinas Scene: The SA dataset was also acquired by the AVIRIS sensor, capturing imagery of the Salinas Valley (California) with a size of 512 × 217. Like the IP dataset, 20 invalid bands were discarded, leaving 204 active bands (spatial resolution: 3.7 m) that cover 16 feature classes. The dataset has a total of 54 129 labeled pixels.

3) Houston 2013: The HU dataset is a typical HSI benchmark for urban remote sensing analysis. Acquired by NASA over the urban area of Houston, Texas, USA, in 2013, it has a dimension of 349 × 1905 with 144 spectral bands. The dataset contains 15 land-cover categories, including residential areas, parks, roads, and buildings, with a total of 15 029 labeled pixels.

4) Fanglu Tea Farm: The FLT dataset was collected from Changzhou's Fanglu Tea Farm (China), including hyperspectral remote sensing images (acquired by the Aerosol Pushbroom Hyperspectral Imager, PHI) and concurrent ground survey data. The image center is at 119°22′53″E, 31°40′39″N, with 80 effective wavelength bands, a spatial size of 348 × 512, and a spatial resolution of 2.25 m. It covers ten feature classes and has a total of 53 734 labeled pixels.

B. Experimental Setting

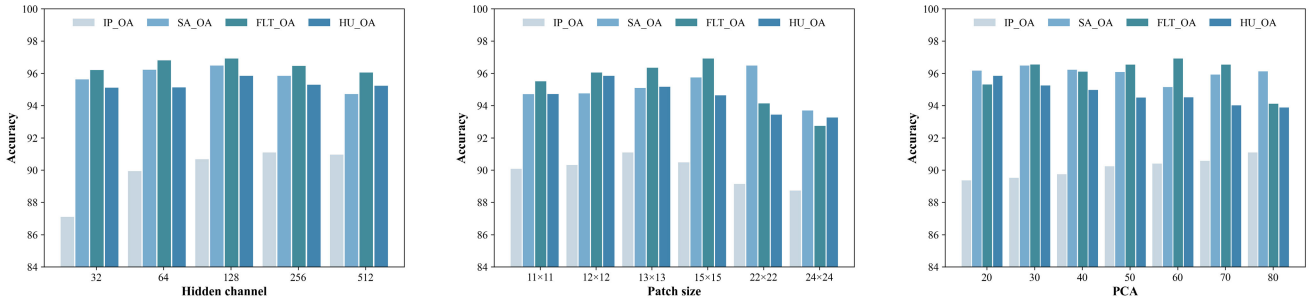
1) Experimental Configuration: All algorithms were run on PyTorch (Intel i7-14650HX CPU, NVIDIA RTX 4060 GPU). Each was repeated ten times (mean/variance as metrics) with

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TABLE I

CLASS DISTRIBUTION AND SAMPLE COUNTS FOR TRAINING AND TESTING ACROSS FOUR DATASETS

NO.	Indian Pines			Salinas Scene			Houston 2013			Fanglu Tea Farm		
	Class	Training.	Test.	Class	Training.	Test.	Class	Training.	Test.	Class	Training.	Test.
1	Alfalfa	15	43	Brocoli_green_weeds_1	15	1994	Healthy grass	30	1221	Masson-pine	20	5786
2	Corn-notill	15	1369	Brocoli_green_weeds_2	15	3711	Stressed grass	30	1224	Bamboo-forest	20	2298
3	Corn-mintill	15	795	Fallow	15	1961	Synthetic grass	30	667	Tea-plant	20	28408
4	Corn	15	226	Fallow-rough-plow	15	1379	Trees	30	1214	Reed	20	196
5	Grass-pasture	15	462	Fallow-smooth	15	2663	Soil	30	1212	Rice-paddy	20	6789
6	Grass-trees	15	699	Stubble	15	3944	Water	30	295	Sweet-potato	20	797
7	Grass-pasture-mowed	15	25	Celery	15	3564	Residential	30	1238	Caraway	20	409
8	Hay-windrowed	15	457	Grapes-untrained	15	11256	Commercial	30	1214	Weed	20	1841
9	Oats	15	17	Soil-yinyard-develop	15	6188	Road	30	1222	Waterbody	20	6121
10	Soybean-notill	15	951	Corn-senesced-green-weeds	15	3263	Highway	30	1197	Building/road	20	891
11	Soybean-mintill	15	2354	Lettuce_romaine_4wk	15	1053	Railway	30	1205			
12	Soybean-clean	15	568	Lettuce_romaine_5wk	15	1912	Parking Lot1	30	1203			
13	wheat	15	194	Lettuce_romaine_6wk	15	901	Parking Lot2	30	439			
14	woods	15	1212	Lettuce_romaine_7wk	15	1055	Tennis court	30	398			
15	Building-Grass-Trees-Drives	15	369	Vinyard-untrained	15	7253	Running track	30	630			
16	Stone-Steel-Towers	15	88	Vinyard-vertical-trellis	15	1792						
-	Total	240	10009	Total	240	53889	Total	450	14579	Total	200	53534



(a) (b) (c)

Fig. 3. Analysis of different parameters. (a) Accuracy of different channel numbers. (b) Accuracy of different patch sizes. (c) Accuracy of different PCAs.

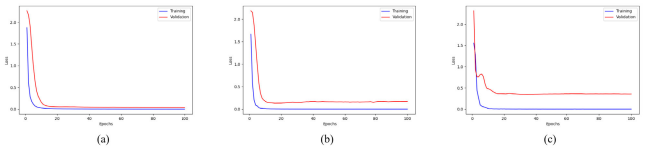
identical train/test datasets for fairness. Our algorithm used the Adam (lr = 0.001 and weight decay = 0.0001), batch size 128, and 300 epochs for convergence across four datasets.

2) Evaluation Metrics: To comprehensively compare the classification performance of different algorithms and explore the parameter influence on experimental results, we adopt both quantitative and qualitative evaluations. Four quantitative metrics are employed in this work: Overall Accuracy (OA), Average Accuracy (AA), Kappa coefficient, and Accuracy per Class (AEC). Specifically, OA represents the proportion of correctly classified samples to all test samples; AA denotes the average classification accuracy over all land-cover classes; the Kappa coefficient quantitatively evaluates the consistency between classification predictions and ground truth; AEC calculates the accuracy of each individual class, which is particularly suitable for evaluating imbalanced datasets.

Besides quantitative evaluation, we also conducted subjective visual evaluation: we visualized the classification results of each algorithm and compared them with real labels. This method lets us visually assess the performance and effectiveness of different algorithms.

C. Parametric Analysis

In the experiment, we thoroughly investigate how parameter settings affect results, focusing on three key parameters: hidden-layer channel size, patch size, and PCA downscaling number. These hyperparameters significantly influenced the overall performance of the model. We systematically analyze classification accuracy under different combinations of parameters in the four datasets to clarify how these parameters influence model performance.



(a) (b) (c)

Fig. 4. Training and validation loss curves for different channel size configurations on the FLT dataset. (a) Channel size is 128. (b) Channel size is 256. (c) Channel size is 512.

To explore how the size of the hidden-layer channel affects the results, we adjusted this parameter while keeping other settings constant. Using a hidden-layer channel size of 128 as the baseline, we tested other sizes, including 32, 64, 256, and 512. As shown in Fig. 3(a), the model accuracy increases across all datasets with the gradual increase in channel size. However, when the channel size is further increased to 256 and 512, the model accuracy begins to show a downward trend. This phenomenon occurs because more channels sharply increase network complexity and the number of parameters to learn. With limited training samples, large-scale networks become harder to train and more prone to overfitting. which impairs the model's generalization ability in practical use. Table II presents the model parameters for different channel

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TABLE II
PARAMETERS (MB) OF OUR MODEL WITH DIFFERENT CHANNEL SIZES ON FOUR DATASETS

Channel Size	IP (MB)	SA (MB)	HU (MB)	FLT (MB)
128	2.13	2.31	1.96	2.15
256	7.96	8.91	7.56	8.13
512	30.74	35.00	29.66	31.59

sizes on all datasets. To further verify the overfitting caused by the increased network complexity and parameter count as the channel size grows, we present the training/validation loss curves under different channel settings. Fig. 4 demonstrates that the gap between the training and validation loss curves widens as the channel size increases, which visually supports the occurrence of overfitting. Experimental data indicate that the model achieves optimal performance on the IP dataset when the hidden-layer channel size is 256, while it attains the highest accuracy on the SA, HU, and FLT datasets when the hidden-layer channel size is set to 128.

As shown in Fig. 3(b), we present the classification results of the four datasets under different patch sizes. Significant differences in the "spectral–spatial" features of different hyperspectral datasets directly determine the optimal patch size. Experimental results demonstrate that the model achieves optimal performance on the IP dataset with a patch size of 13×13; it reaches the highest accuracy on the SA dataset when using a 22×22 patch size; a 12×12 patch size enables the model to attain optimal performance on the HU dataset; and the model delivers the best performance on the FLT dataset with a patch size of 15×15.

Our model uses PCA, so different PCA downscaling numbers directly affect classification accuracy across datasets. Fig. 3(c) shows the overall classification accuracy (OA) of the four hyperspectral datasets (IP, SA, HU, and FLT) under different PCA dimensions (20, 30, 40, 50, 60, 70, and 80). Overall, PCA dimensions impact the datasets' classification accuracy differently. The model achieves optimal performance on the IP dataset with 80 PCA dimensions and attains the highest accuracy on the SA dataset with 30 PCA dimensions. Similarly, it delivers the best performance on the HU and FLT datasets using 20 and 60 PCA dimensions, respectively.

D. Ablation Study

To evaluate the contribution of each module in our proposed PGC-SCTN model, we design comprehensive ablation experiments on four datasets, keeping parameter settings consistent throughout. The complete PGC-SCTN model comprises three core modules: DCE, PGFE, and SCT. In this section, we first conduct a comprehensive ablation experiment on each major module of the PGC-SCTN model to evaluate the effectiveness of these modules. It is worth noting that all ablation experiments are the average results of ten randomized experiments. Table III presents the performance metrics of each ablation experiment.

1) Ablation Experiments on the Major Modules: We establish the baseline by removing all major modules and adopting a traditional convolutional layer (comprising 2-D convolution, BN, and GeLU). First, we analyze each major module individually. Experimental comparisons show that using the SCT module alone improves performance by 2.61%, 2.41%, 4.29%, and 6.77% on the IP, SA, HU, and FLT datasets, respectively. On these four datasets, the standalone PGFE module boosts performance by 3.26%, 2.46%, 3.33%, and 6.63%, while the individual DCE module enhances performance by 3.61%, 2.30%, 3.90%, and 6.53%, respectively. Experimental results demonstrate that each proposed module contributes to the model's performance improvement.

We then analyze the experimental results of pairwise combinations of the major modules. On different datasets, any combination of two modules exerts varying degrees of impact on performance. Experimental results show that, with few exceptions, combining any two modules typically yields higher accuracy than using a single module. For some combinations where performance is slightly lower than that of individual modules, it can be explained by the mismatched feature dimensions and expression logics of the outputs from different modules—the model struggles to fuse the two types of features efficiently, which instead dilutes the advantages of the individual modules. However, all pairwise module combinations still outperform the baseline, confirming that each proposed module effectively improves accuracy. Finally, we compare the model with all major modules enabled against the baseline and find that the model incorporating all major modules achieves the highest accuracy across all datasets, with improvements of 4.62%, 2.66%, 4.40%, and 7.41%, respectively. This further demonstrates the significant advantages of our proposed major modules in enhancing accuracy.

2) Ablation Experiments on the PGC Module: To verify the effectiveness of the PGC we developed, we designed an additional ablation experiment where the PGC module is replaced with a similar standard 2-D convolution layer, and the model is trained for comparison. As shown in Table IV, when using the PGFE major module alone, enabling the PGC module achieves performance improvements of 0.70%, 0.16%, 0.14%, and 0.56% on the IP, SA, HU, and FLT datasets, respectively, compared with replacing it with a 2-D convolution layer. Similarly, when all major modules are enabled, the model with the PGC module outperforms the one with the replaced layer. These results further confirm the contribution of our proposed PGC module to the feature extraction capability of PGC-SCTN.

3) Ablation Experiments on the SCT: To further verify the effectiveness of the proposed SCT module in the entire network, we first train and compare the model using only the original Transformer encoder (TE). As shown in Table V, enabling the SCT module alone achieves performance improvements of 1.11%, 0.27%, 1.41%, and 3.04% across all datasets compared with using the TE alone. Similarly, we compare the overall networks fused with the SCT module and the TE module, respectively, and the former delivers superior model performance. Experimental results demonstrate that our proposed SCT module better captures the global spectral–spatial semantic information of HSI and establishes long-range dependencies.

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TABLE III

RESULTS OF THE ABLATION STUDIES ON THE MAJOR MODULES CONDUCTED ON THE IP, SA, HU, AND FLT DATASETS ARE SHOWN BELOW (THE BEST RESULTS ARE IN BOLD)

Module			Indian Pine			Salinas Scene			Houston 2013			Fanglu Tea Farm		
DCE	PGFE	SCT	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
			86.47±1.48	92.53±0.67	84.70±1.65	93.82±0.93	96.44±0.51	93.14±1.03	91.44±1.09	92.83±0.93	90.74±1.18	89.66±2.00	95.33±0.56	85.66±2.59
		√	89.08±0.98	93.67±0.45	87.61±1.08	96.23±1.14	97.97±0.50	95.80±1.27	95.73±1.04	96.45±0.88	95.38±1.12	96.43±0.96	97.18±0.71	94.81±1.37
	√		89.73±0.75	93.83±0.50	88.33±0.84	96.28±1.23	97.84±0.53	95.86±1.37	94.77±1.37	95.58±1.17	94.34±1.48	96.29±0.76	97.16±0.55	94.61±1.09
√			90.08±1.02	94.09±0.35	88.74±1.14	96.12±1.23	97.85±0.54	95.70±1.37	95.34±0.82	96.05±0.72	94.96±0.89	96.19±1.10	97.12±0.44	94.46±1.56
	√	√	89.89±1.01	94.13±0.36	88.52±1.13	96.42±1.15	98.00±0.54	96.02±1.28	95.66±0.69	96.39±0.57	95.31±0.74	96.65±0.73	97.34±0.56	95.11±1.05
√		√	90.08±1.16	94.29±0.38	88.75±1.30	96.13±1.22	98.10±0.45	95.70±1.36	95.70±0.45	96.39±0.40	95.35±0.49	95.74±1.64	97.02±0.53	93.85±2.28
√	√		89.96±1.11	94.07±0.43	88.60±1.24	96.12±1.14	97.89±0.58	95.68±1.56	94.94±0.74	95.78±0.58	94.53±0.80	96.22±1.08	97.03±0.58	94.51±1.51
√	√	√	**91.09±0.89**	**94.80±0.40**	**89.87±1.02**	**96.48±1.00**	**98.26±0.36**	**96.09±1.11**	**95.84±0.76**	**96.47±0.67**	**95.50±0.82**	**97.07±0.63**	**97.56±0.39**	**95.72±0.90**

TABLE IV

ABLATION EXPERIMENT RESULTS OF THE PGC ON DIFFERENT DATASETS

Modules				Datasets			
DCE	PGFE		SCT	IP	SA	HU	FLT
	Convd	PGC					
	√			89.03±0.95	96.12±1.29	94.63±1.14	95.73±0.93
		√		**89.73±0.75**	**96.28±1.23**	**94.77±1.37**	**96.29±0.76**
√	√		√	90.13±1.17	94.92±4.31	95.25±1.01	95.97±1.45
√		√	√	**91.09±0.89**	**96.48±1.00**	**95.84±0.76**	**97.07±0.63**

TABLE V

ABLATION EXPERIMENT RESULTS OF THE SCT ON DIFFERENT DATASETS

Module				Datasets			
DCE	PGFE	SCT	TE	IP	SA	HU	FLT
			√	87.97±1.42	95.96±1.34	94.32±0.90	93.39±1.82
		√		**89.08±0.98**	**96.23±1.14**	**95.73±1.04**	**96.43±0.96**
√	√		√	90.76±1.06	96.20±1.23	94.45±1.55	96.72±1.46
√	√	√		**91.09±0.89**	**96.48±1.00**	**95.84±0.76**	**97.07±0.63**

E. Comparative Experiments

To evaluate the effectiveness of our proposed algorithm, we compare it with eight SOTA algorithms. The CNN-based algorithms include SPRN, SSEFN, and CSCAN [41], while the CNN–Transformer hybrid algorithms consist of SSFTT, GSCVit [42], and CLMG [43]. Furthermore, to fully verify the superiority of PGC-SCTN for few-shot HSI classification in modeling global spectral–spatial dependencies, we also adopt two advanced HSI classification methods designed for global correlation modeling as baselines: NSCKL [44] and MS2FN [45]. Notably, SSEFN is specifically designed for HSI classification with limited labeled samples. For a fair comparison, we adopt the parameter settings recommended in the original papers of these comparative methods.

1) Quantitative Accuracy Analysis: Tables VI–IX present the quantitative results, where the results of all comparison experiments are presented as mean ± standard deviation, with the optimal result in each row marked in bold. Observing these tables, it can clearly see that our proposed PGC-SCTN achieves the best performance across all four datasets. Furthermore, compared with other methods, the proposed approach exhibits the smallest standard deviation after ten experiments.

Table VI presents the classification results of each algorithm on the IP dataset. As shown in the table, our algorithm outperforms other methods in most land-cover categories, demonstrating its effectiveness. However, for certain categories, other algorithms can achieve comparable or even better performance. The explanation is that the large number of feature classes in the IP dataset and the uneven sample distribution between some classes. We observe that MS2FN achieves a promising classification accuracy of 88.09%, which is higher than that of other CNN-based and CNN–Transformer-based methods. This arises because MS2FN constructs pixel-level graph structures from multiscale segmented data, thereby enabling effective modeling of the global spectral–spatial features in HSI. Notably, the proposed algorithm achieves the best OA of 91.09%.

Table VII presents the classification results of various algorithms on the SA dataset. Our algorithm demonstrates distinct advantages across the three evaluation metrics and most land-cover categories, achieving a maximum OA of 96.48%. Tables VIII and IX show the classification results on the HU and FLT datasets, respectively. Similarly, our algorithm attains high OA and superior performance in most feature classes. Notably, SSEFN, an algorithm specifically designed for few-shot HSI classification, performs relatively poorly on the IP, SA, and FLT datasets. The explanation is that SSEFN's primary optimization for training with extremely limited samples. Our experiments exceed its optimal adaptation range, preventing its core mechanisms from functioning effectively. Furthermore, given the characteristics of different datasets—such as high dimensionality, unbalanced distribution of land-cover classes, and high redundancy of local features—SSEFN also exhibits relatively low accuracy across these datasets, indicating insufficient feature representation and generalization capabilities of the model itself. However, our algorithm consistently delivers the best classification performance under diverse conditions, confirming its superiority.

2) Qualitative Analysis: For a more objective and rigorous evaluation of the model's generalization ability on the entire hyperspectral scene, we generate full-scene pixel-level classification maps for all comparative methods and the proposed PGC-SCTN, covering every pixel in the original HSI scene. We present the classification result maps of the four datasets in Figs. 5–8.

Overall, compared with other methods, our proposed PGC-SCTN model produces fewer misclassifications and generates high-quality classification maps with smooth objects and well-preserved boundaries. Fig. 6 shows the classification maps

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TABLE VI
CLASSIFICATION RESULTS(%) ON IP DATASET WITH 15 LABELED SAMPLE EACH CLASS

Class	SPRN(2022)	SSFTT(2022)	NSCKL(2023)	GSCVit(2024)	CLMG(2024)	MS2FN(2025)	SSEFN(2025)	CSCAN(2025)	PGC-SCTN
1	99.68±0.97	**100.00±0.00**	82.90±9.24	99.68±0.97	**100.00±0.00**	**100.00±0.00**	98.39±2.60	99.68±0.97	**100.00±0.00**
2	77.33±26.81	77.91±6.73	79.03±6.02	79.33±12.24	79.52±5.51	83.51±7.02	56.78±7.62	77.02±5.77	**87.10±5.81**
3	86.70±8.14	82.53±6.38	72.72±6.44	83.79±11.45	84.99±5.68	84.87±7.97	68.69±6.78	82.47±5.26	**87.80±3.87**
4	98.88±1.59	97.39±3.50	65.58±14.46	96.35±3.63	**98.92±1.62**	98.11±2.99	93.92±4.47	96.80±2.92	97.03±2.67
5	84.79±20.03	89.62±3.53	90.91±2.30	87.50±6.13	89.57±3.84	**91.71±3.79**	87.84±2.76	90.59±3.81	90.60±3.47
6	98.88±1.30	98.15±2.41	97.10±1.84	98.26±1.20	98.45±0.84	**98.99±1.01**	94.18±4.20	97.75±2.61	98.03±1.38
7	**100.00±0.00**	**100.00±0.00**	96.15±6.20	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	98.19±5.44	99.59±1.09	99.13±1.08	99.91±0.20	99.96±0.09	99.83±0.48	**100.00±0.00**	99.61±0.80	99.85±0.26
9	**100.00±0.00**	**100.00±0.00**	98.00±6.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	81.07±13.07	84.85±4.15	84.85±3.68	77.96±14.99	84.02±6.27	83.62±7.98	70.10±8.00	84.06±5.76	**86.69±5.51**
11	71.36±21.56	79.36±6.45	86.52±4.05	70.44±14.58	74.88±5.69	79.16±8.17	63.27±10.88	80.62±4.18	**87.11±3.08**
12	88.60±16.42	80.16±4.97	77.84±7.92	81.26±11.04	79.41±8.07	84.97±7.32	64.98±9.53	83.56±4.45	**88.79±2.55**
13	99.95±0.16	**100.00±0.00**	98.72±0.70	99.21±1.48	99.89±0.21	99.74±0.51	99.47±0.58	99.16±1.56	99.53±0.93
14	89.75±17.18	97.58±3.01	96.36±1.94	97.02±3.41	94.27±4.10	**98.31±1.46**	95.69±2.11	97.59±3.26	97.69±3.16
15	96.31±4.05	96.77±4.03	66.91±7.90	91.75±12.45	94.34±4.68	97.12±5.27	86.79±6.70	95.15±5.30	**97.81±1.81**
16	99.87±0.38	99.36±1.92	92.87±3.22	**99.87±0.38**	**99.87±0.38**	99.74±0.54	97.56±3.65	98.21±2.51	98.72±1.81
OA	86.82±10.74	86.72±1.79	85.31±1.06	83.89±6.34	85.50±1.08	88.09±2.45	75.35±2.44	86.94±1.30	**91.09±0.89**
AA	92.96±4.78	92.70±0.65	82.26±1.69	91.40±3.50	92.38±0.86	93.73±1.39	86.10±1.37	92.64±0.80	**94.80±0.40**
Kappa	85.15±11.69	84.94±1.98	83.21±1.22	81.83±7.04	83.61±1.19	86.47±2.77	72.16±2.66	85.18±1.44	**89.87±1.02**

TABLE VII
CLASSIFICATION RESULTS(%) ON SA DATASET WITH 15 LABELED SAMPLES EACH CLASS

Class	SPRN(2022)	SSFTT(2022)	NSCKL(2023)	GSCVit(2024)	CLMG(2024)	MS2FN(2025)	SSEFN(2025)	CSCAN(2025)	PGC-SCTN
1	99.70±0.77	99.92±0.21	98.96±1.18	99.37±1.74	99.84±0.28	99.83±0.54	99.80±0.41	**99.97±0.05**	99.83±0.26
2	99.94±0.18	99.42±1.10	99.48±1.16	87.84±29.55	99.32±1.04	**100.00±0.00**	99.84±0.15	99.92±0.15	99.99±0.03
3	98.82±3.55	99.62±0.88	98.58±2.57	96.33±10.85	99.84±0.23	99.57±1.10	**100.00±0.00**	99.93±0.12	**100.00±0.00**
4	99.62±0.78	97.50±5.02	**99.78±0.12**	99.71±0.36	99.52±0.41	99.75±0.21	99.55±0.34	99.26±0.73	99.23±1.01
5	97.98±1.70	96.13±2.89	98.56±1.01	94.49±6.76	96.23±2.22	97.80±1.80	**99.25±0.43**	95.50±3.25	96.97±2.45
6	**99.94±0.17**	99.00±1.95	99.85±0.07	99.89±0.17	**99.94±0.14**	99.92±0.06	99.91±0.09	99.29±0.95	97.87±2.46
7	99.90±0.12	99.28±1.25	99.79±0.10	92.78±19.67	99.19±0.77	99.77±0.18	**99.93±0.03**	99.86±0.23	99.24±1.20
8	78.05±13.77	80.28±7.91	75.04±11.98	67.09±34.11	76.43±6.95	64.28±17.49	76.91±5.58	83.42±4.12	**89.00±4.48**
9	99.83±0.42	99.86±0.21	99.54±0.41	99.84±0.47	99.96±0.08	99.99±0.03	**100.00±0.00**	99.97±0.05	**100.00±0.00**
10	94.01±1.82	92.78±3.01	94.88±0.92	95.42±2.02	94.40±2.19	95.99±2.20	94.69±1.91	95.98±1.29	**97.69±1.36**
11	99.61±0.55	99.81±0.43	98.47±1.09	99.80±0.22	99.87±0.22	98.17±2.63	98.54±1.46	**99.97±0.06**	99.83±0.33
12	99.94±0.05	95.05±7.70	99.83±0.10	99.33±0.67	99.73±0.22	**99.95±0.05**	99.83±0.28	99.76±0.24	98.74±1.16
13	99.38±0.65	96.90±7.00	99.48±0.30	**99.90±0.20**	99.89±0.17	99.49±0.47	99.78±0.37	99.69±0.72	99.66±0.39
14	99.78±0.29	99.15±1.38	98.34±0.92	99.46±0.74	99.78±0.28	99.14±0.85	95.23±4.60	**99.94±0.10**	98.98±1.04
15	87.60±9.68	89.11±8.87	73.42±11.27	76.72±24.49	88.57±4.97	86.32±11.41	75.12±9.57	92.75±7.44	**95.44±5.42**
16	98.93±1.28	98.67±0.71	99.04±0.50	98.28±2.34	98.64±2.44	98.59±0.50	98.94±0.70	99.26±0.69	**99.63±0.64**
OA	93.12±2.17	93.23±1.54	90.20±1.28	87.84±6.55	92.82±1.46	90.19±2.88	91.26±1.23	94.96±0.56	**96.48±1.00**
AA	97.06±0.70	96.41±1.12	95.81±0.31	94.14±4.13	96.95±0.54	96.16±0.96	96.08±0.71	97.78±0.27	**98.26±0.36**
Kappa	92.37±2.38	92.50±1.69	89.05±1.40	86.56±7.16	92.03±1.62	89.13±3.15	90.28±1.37	94.40±0.62	**96.09±1.11**

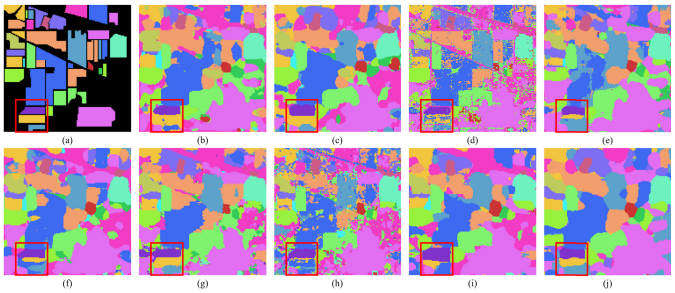


Fig. 5. Classification map of the IP dataset. (a) Ground-truth map. (b) SPRN. (c) SSFTT. (d) NSCKL. (e) GSCVit. (f) CLMG. (g) MS2FN. (h) SSEFN. (i) CSCAN. (j) PGC-SCTN.

of all compared algorithms on the SA dataset. We observe the upper left quadrant of the classification maps: all other models contain numerous noisy regions, misclassified areas, and blurred boundaries. In contrast, the PGC-SCTN model exhibits better target continuity and clearer boundaries. Similarly, as shown in Fig. 7, when examining the upper middle region of the classification map, we observe that our algorithm produces fewer misclassifications and generates clearer results than other methods. For all datasets, we use the regions marked with red boxes as examples. These demonstrate that our model's classification maps have less noise than those of other methods and are closer to the true distribution.

To further compare the performance of PGC-SCTN and the comparative models in an intuitive manner, we adopt t-distributed stochastic neighbor embedding (t-SNE) to visually project the features onto a 2-D coordinate system. Fig. 9 presents the 2-D plot on the IP dataset, where points of different colors represent samples of different categories. The

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TABLE VIII
CLASSIFICATION RESULTS(%) ON HU DATASET WITH 30 LABELED SAMPLES PER CLASS

Class	SPRN(2022)	SSFTT(2022)	NSCKL(2023)	GSCVit(2024)	CLMG(2024)	MS2FN(2025)	SSEFN(2025)	CSCAN(2025)	PGC-SCTN
1	91.96±5.73	93.54±4.49	94.42±5.33	91.80±5.45	93.55±4.66	93.98±5.98	**94.31±4.38**	91.33±5.23	94.00±4.07
2	93.86±4.92	94.88±2.60	91.89±4.42	90.60±5.30	97.07±2.07	92.79±9.27	96.36±1.73	97.17±2.81	**97.41±1.33**
3	95.56±11.58	99.01±0.81	99.34±0.92	74.92±36.72	99.62±0.33	99.61±0.70	**100.00±0.00**	99.57±0.42	99.31±0.76
4	93.13±10.53	95.41±3.02	92.91±3.37	93.72±0.58	95.13±2.88	94.48±1.55	**98.45±2.32**	96.86±3.06	97.01±2.60
5	99.77±0.47	99.37±1.25	97.65±1.33	97.12±5.55	98.62±3.33	98.93±1.08	99.88±0.07	99.94±0.12	**100.00±0.00**
6	98.51±1.89	96.95±6.16	98.80±1.03	94.88±5.38	98.14±1.94	97.08±3.63	**99.39±0.93**	99.25±1.67	98.61±3.00
7	85.66±10.16	87.11±4.22	85.04±4.60	88.56±5.14	91.47±3.60	89.21±5.99	**94.52±3.09**	91.30±2.93	93.71±2.55
8	75.32±8.02	80.57±5.35	81.31±5.56	78.90±7.18	85.02±4.65	80.39±7.49	83.38±2.63	85.39±5.21	**88.81±3.54**
9	87.01±5.09	78.94±6.04	79.17±7.22	85.02±11.62	88.87±3.43	84.13±6.26	90.02±4.00	84.40±11.91	**90.23±2.75**
10	90.87±10.66	92.12±11.77	87.36±4.02	82.92±17.23	96.42±2.28	83.55±10.17	**98.94±0.92**	94.84±5.24	98.38±1.48
11	89.20±6.11	95.67±4.63	79.76±3.89	79.62±13.94	91.92±3.17	85.05±6.07	**98.63±1.37**	96.16±4.05	98.59±3.02
12	85.74±5.96	87.01±6.84	79.02±5.09	71.56±18.09	91.12±3.72	92.03±6.53	89.28±5.69	89.99±4.70	**93.78±3.37**
13	92.39±3.07	93.03±3.75	69.95±4.91	93.39±4.30	**98.20±2.13**	92.98±4.56	85.90±3.65	94.24±3.96	97.38±2.20
14	**100.00±0.00**	99.97±0.08	99.50±0.52	99.72±0.43	**100.00±0.00**	99.70±0.41	**100.00±0.00**	99.97±0.08	99.92±0.16
15	**100.00±0.00**	**100.00±0.00**	99.00±0.57	99.89±0.33	99.95±0.14	99.86±0.25	99.65±0.27	**100.00±0.00**	99.95±0.10
OA	90.58±1.82	91.72±0.92	88.02±0.89	86.88±4.08	93.98±0.55	90.92±1.81	94.86±0.83	93.73±1.26	**95.84±0.76**
AA	91.93±1.68	92.91±0.84	89.01±0.67	88.18±4.26	95.01±0.55	92.25±1.42	95.25±0.75	94.70±1.12	**96.47±0.67**
Kappa	89.81±1.97	91.05±0.99	87.04±0.96	85.82±4.40	93.49±0.60	90.18±1.96	94.44±0.90	93.22±1.37	**95.50±0.82**

TABLE IX
CLASSIFICATION RESULTS(%) ON FLT DATASET WITH 20 LABELED SAMPLES PER CLASS

Class	SPRN(2022)	SSFTT(2022)	NSCKL(2023)	GSCVit(2024)	CLMG(2024)	MS2FN(2025)	SSEFN(2025)	CSCAN(2025)	PGC-SCTN
1	90.73±10.02	98.10±1.37	89.82±2.69	86.36±20.44	96.03±2.20	99.26±1.04	98.64±1.54	98.43±1.07	**99.64±0.60**
2	83.30±10.70	89.84±5.35	69.39±2.88	93.82±5.42	94.42±3.00	91.66±7.63	72.34±6.49	91.41±4.05	**97.28±2.11**
3	91.50±3.13	87.26±9.52	88.06±2.16	92.70±2.98	94.08±1.95	86.00±7.08	89.74±3.37	93.44±2.68	**95.97±1.21**
4	**100.00±0.00**	99.90±0.31	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
5	99.21±1.63	99.11±0.41	97.95±2.58	92.79±17.58	99.62±0.36	**99.86±0.23**	99.81±0.21	99.43±0.41	98.97±0.45
6	94.69±4.09	95.20±2.66	**98.10±1.16**	92.89±4.45	97.13±2.12	96.37±2.58	95.71±1.81	96.09±1.27	96.00±1.68
7	99.93±0.22	99.78±0.35	98.86±0.74	99.66±0.73	**100.00±0.00**	99.93±0.23	99.54±0.95	**100.00±0.00**	**100.00±0.00**
8	93.90±4.40	85.77±8.99	84.97±3.04	90.10±11.32	**95.42±5.24**	90.80±8.44	78.16±8.96	92.96±4.61	92.56±4.30
9	99.25±0.54	99.20±0.57	99.31±0.29	99.06±0.45	99.04±0.48	99.06±0.46	99.17±0.65	**99.53±0.41**	98.87±0.33
10	98.64±2.01	96.66±3.68	99.55±0.71	86.16±25.50	99.11±0.76	**99.55±0.94**	98.59±0.64	96.48±2.55	96.34±2.54
OA	93.27±1.78	91.78±5.26	90.27±1.16	92.69±3.49	95.81±1.16	91.63±3.65	92.26±1.74	95.50±1.32	**97.07±0.63**
AA	95.11±1.22	95.08±1.90	92.60±0.44	93.35±4.24	97.48±0.67	96.25±1.56	93.17±0.86	96.79±0.49	**97.56±0.39**
Kappa	90.35±2.45	88.60±6.55	85.91±1.56	89.50±4.93	93.94±1.64	88.33±4.84	88.94±2.34	93.50±1.83	**95.72±0.90**

TABLE X
COMPARING THE MODELING EFFICIENCY OF THE DIFFERENT METHODS ON THE IP, SA, HU, AND FLT DATASETS

Dataset	metrics	SPRN	SSFTT	NSCKL	GSCVit	CLMG	MS2FN	SSEFN	CSCAN	PGC-SCTN
IP	$T_{\text{train}}(S)$	0.05	0.03	**0.01**	0.03	0.32	0.30	1.12	0.06	0.13
	$T_{\text{test}}(S)$	0.73	0.34	**0.02**	0.42	3.76	13.96	7.06	0.63	1.88
	Params(M)	0.73	0.59	**0.12**	0.42	15.12	1.80	41.04	1.06	7.96
	FLOPs(M)	9.04	11.41	0.23	10.17	473.65	**0.11**	2679.12	25.39	335.95
SA	$T_{\text{train}}(S)$	0.05	0.03	**0.01**	0.05	0.36	0.18	1.16	0.06	0.19
	$T_{\text{test}}(S)$	2.22	1.06	**0.05**	6.11	23.71	88.39	40.23	3.42	14.46
	Params(M)	0.73	0.59	**0.12**	0.38	15.24	1.79	41.04	1.06	2.31
	FLOPs(M)	9.04	11.41	0.23	13.19	474.76	**0.09**	2679.15	25.39	278.78
HU	$T_{\text{train}}(S)$	0.08	0.06	**0.02**	0.05	0.61	0.54	2.16	0.12	0.09
	$T_{\text{test}}(S)$	0.55	0.44	**0.34**	0.61	5.78	68.89	10.45	0.94	0.79
	Params(M)	0.72	0.59	**0.10**	0.31	14.67	1.78	41.02	1.05	1.96
	FLOPs(M)	8.94	11.41	0.20	7.62	458.02	**0.09**	2679.14	25.39	70.32
FLT	$T_{\text{train}}(S)$	0.05	0.03	**0.01**	0.02	0.28	0.22	1.11	0.06	0.06
	$T_{\text{test}}(S)$	2.32	1.51	**0.04**	1.77	23.52	287.20	44.32	3.14	4.88
	Params(M)	0.71	0.59	**0.08**	0.25	14.06	1.75	40.92	1.04	2.15
	FLOPs(M)	8.82	11.41	0.17	8.62	440.6	**0.09**	2679.12	25.39	120.58

clustering of samples within the same color indicates the effectiveness of the method in category discrimination. To avoid unnecessary interference, we exclude the background category. PGC-SCTN yields superior clustering results, with only a small number of misclassified samples in the plot. In contrast, the sample distributions of other methods are

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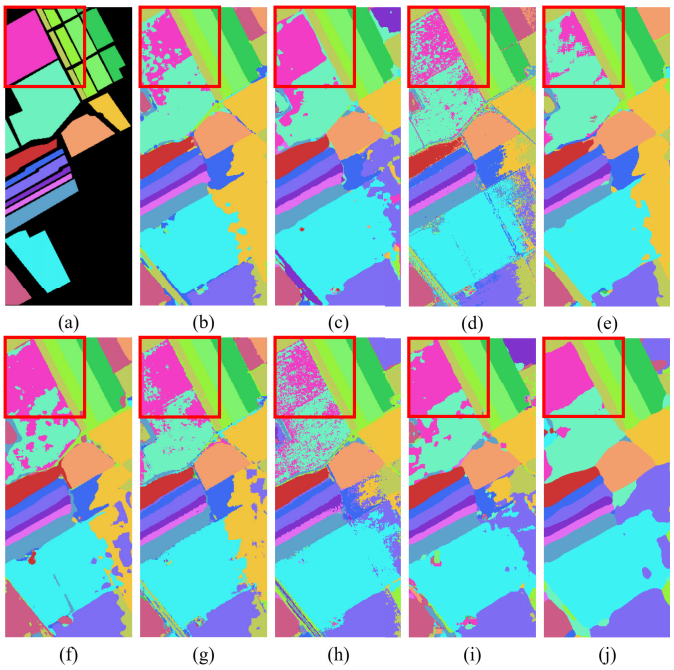


Fig. 6. Classification map of the SA dataset. (a) Ground-truth map. (b) SPRN. (c) SSFTT. (d) NSCKL. (e) GSCVit. (f) CLMG. (g) MS2FN. (h) SSEFN. (i) CSCAN. (j) PGC-SCTN.

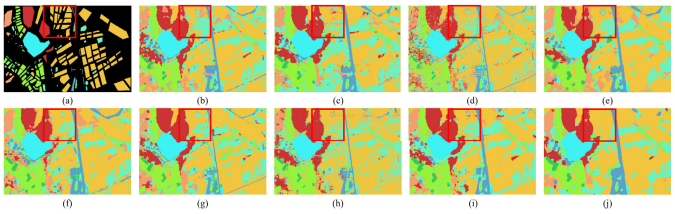


Fig. 7. Classification map of the FLT dataset. (a) Ground-truth map. (b) SPRN. (c) SSFTT. (d) NSCKL. (e) GSCVit. (f) CLMG. (g) MS2FN. (h) SSEFN. (i) CSCAN. (j) PGC-SCTN.

relatively scattered and exhibit more misclassifications. This further demonstrates that our PGC-SCTN achieves excellent classification performance.

F. Computational Complexity and Efficiency Analysis

We conducted a comparative study on the efficiency of various methods and summarized the results in Table X. This table covers indicators such as training time, testing time, model parameters, and computational cost, and intuitively presents the differences in efficiency levels among the methods. Notably, we performed ten random experiments for each method to obtain the average training time and average testing time. Furthermore, due to the sample size, the training process typically runs faster than the testing process.

In terms of model parameters and computational complexity, PGC-SCTN falls into the lightweight category with a parameter count ranging from 1.96 to 7.96 M, which is significantly lower than that of models such as SSEFN (40.02–41.04 M) and CLMG (14.06–15.24 M). Its computational complexity (70.32–335.95 M FLOPs) accounts for only 1/8 to 1/38 of SSEFN's and is reduced by approximately 25%–30% compared with CLMG (440.6–474.76 M FLOPs), substantially lowering the theoretical computational overhead. Regarding training and testing time, our model ranks in the middle across all four datasets. Overall, based on its lightweight parameter scale and low computational complexity, PGC-SCTN achieves efficient training and testing speeds while maintaining leading classification accuracy.

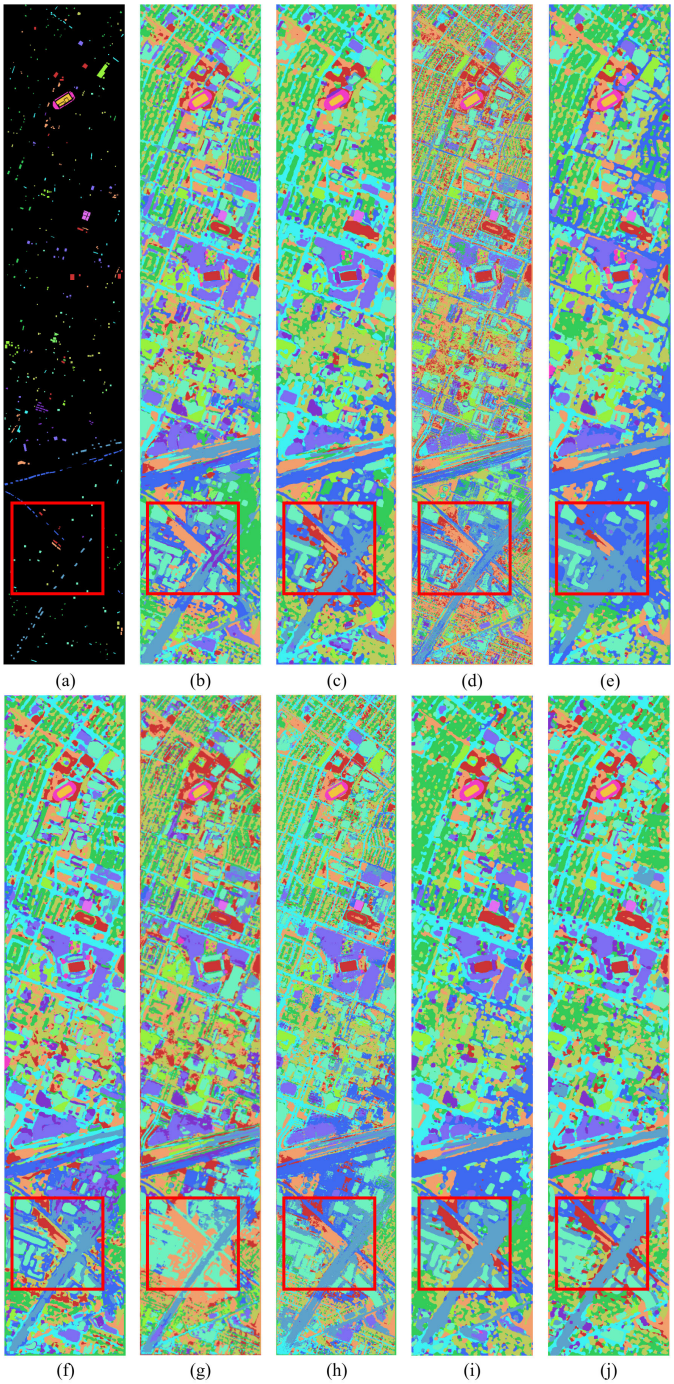


Fig. 8. Classification map of the HU dataset. (a) Ground-truth map. (b) SPRN. (c) SSFTT. (d) NSCKL. (e) GSCVit. (f) CLMG. (g) MS2FN. (h) SSEFN. (i) CSCAN. (j) PGC-SCTN.

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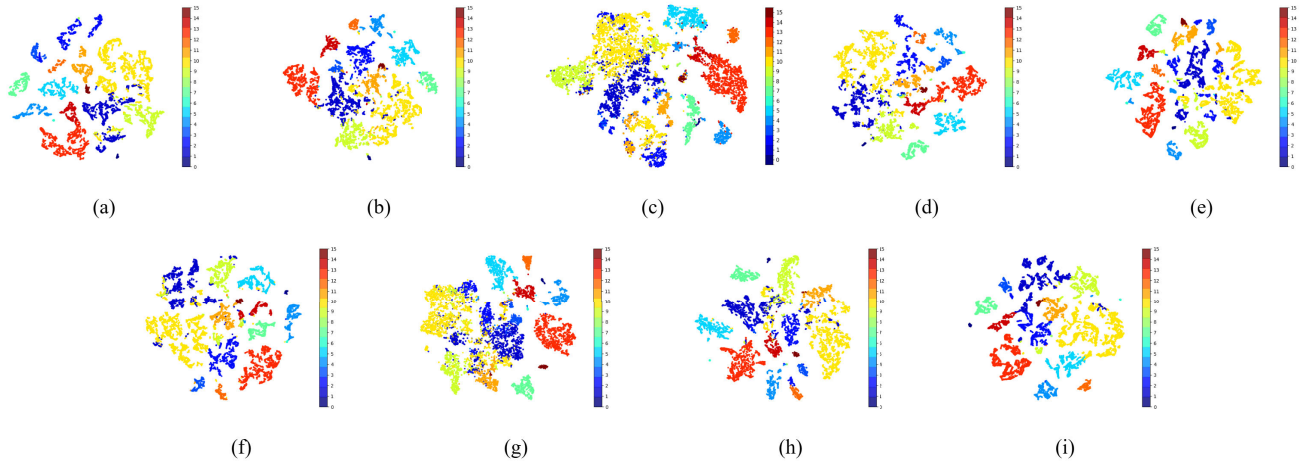


Fig. 9. T-SNE maps of different methods on the India Pines dataset. (a) SPRN (OA = 86.82%). (b) SSFTT (OA = 86.72%). (c) NSCKL (OA = 85.31%). (d) GSCVit (OA = 83.89%). (e) CLMG (OA = 85.50%). (f) MS2FN (OA = 88.09%). (g) SSEFN (OA = 75.35%). (h) CSCAN (OA = 86.94%). (i) PGC-SCTN (OA = 91.09%).

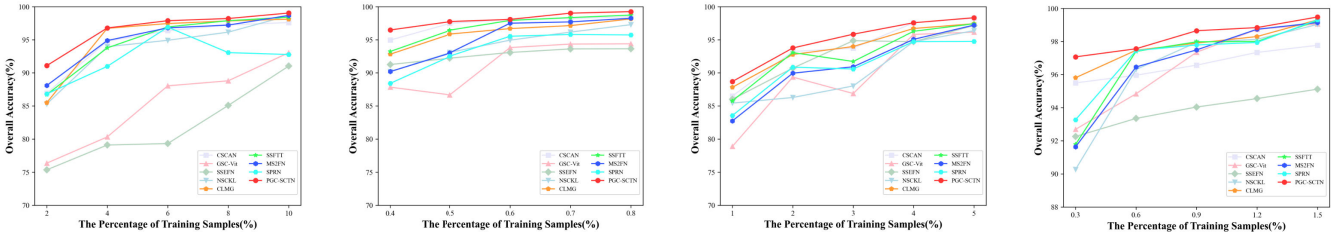


Fig. 10. Classification accuracy of different sample proportions. (a) IP. (b) SA. (c) HU. (d) FLT.

G. Different Proportions of Training Samples

Pixel labeling of HSI is costly, leading to scarce training samples. Earlier experiments showed our proposed algorithm performs optimally with limited data. Here, we test its performance against eight other models on four datasets under different training sample proportions: 2%, 4%, 6%, 8%, and 10% for Indian Pines; 0.4%, 0.5%, 0.6%, 0.7%, and 0.8% for Salinas; 1%, 2%, 3%, 4%, and 5% for Houston 2013; and 0.3%, 0.6%, 0.9%, 1.2%, and 1.5% for Fanglu Tea Farm. The classification accuracy of each model is shown in Fig. 10.

Experimental results show that all algorithms perform better with a larger proportion of training samples and reach optimal performance at the maximum proportion (10% for Indian Pines, 0.8% for Salinas, 5% for Houston 2013, and 1.5% for Fanglu Tea Farm). Notably, our algorithm consistently outperforms others. As the number of training samples decreases, all algorithms' performance declines to varying degrees. In contrast, our algorithm still maintains excellent performance with fewer samples, demonstrating its robustness across different training sample sizes.

H. Experiments on a Challenging Modern HSI Dataset

To verify the generalization and robustness of the proposed PGC-SCTN on recent and challenging HSI classification scenarios, we adopt the Liao Ning 01 (LN01) dataset [46] and randomly select 20 samples per class as the training set for our experiments. The experimental settings, including network parameters, optimizer, and evaluation metrics, are kept strictly consistent with those used in the main experiments to ensure fairness and comparability.

TABLE XI
NUMBER OF TRAINING AND TEST SAMPLES ON THE LIAO NING 01

No.	Class	Training	Test
1	Reservoir	20	5456
2	Seawater	20	100256
3	Sandy soil	20	14567
4	Broken bridge	20	514
5	Barren grass	20	16479
6	Highway	20	5563
7	Railway	20	1410
8	Bare soil	20	6267
9	Mountain vegetation	20	95051
10	Arable land	20	19241
	Total	200	264804

LN01 was captured by the ZY1-02D satellite in the northwestern part of Jinpu New District, Dalian City, Liaoning Province, in September 2021. The central coordinates are 39.98°N, 121.49°E. The dataset consists of 900 × 900 pixels with a spatial resolution of 30 m and 166 spectral bands. It contains ten land-cover classes with a total of 265 004 labeled samples. The numbers of training and test samples for LN01 are summarized in Table XI, the classification results are

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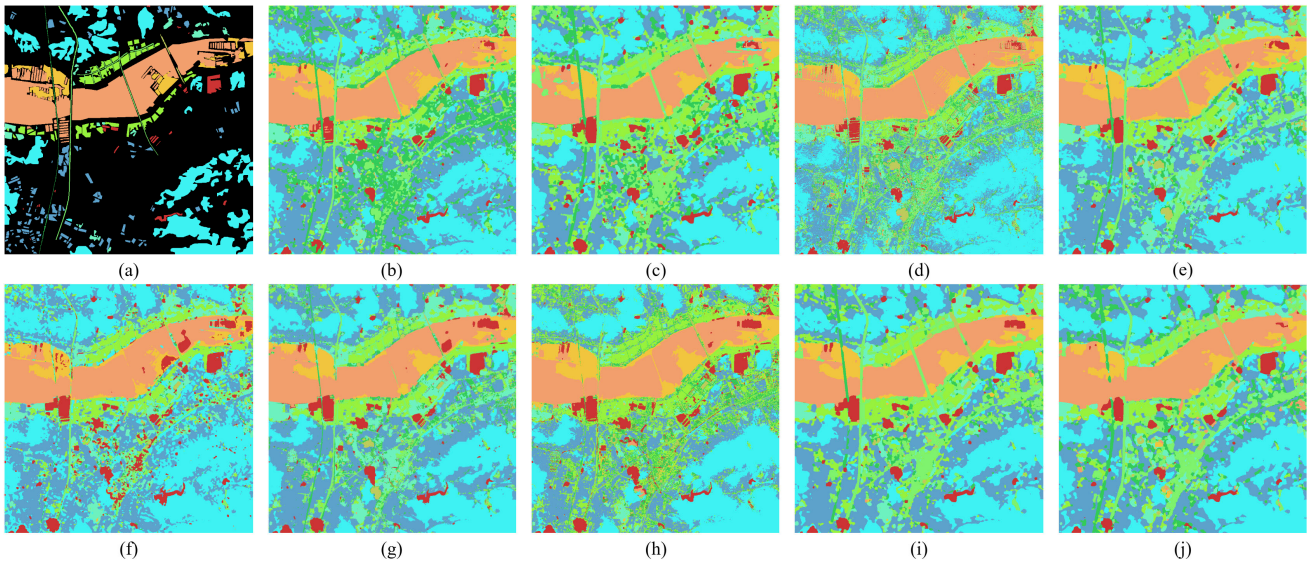


Fig. 11. Classification map of the LN01 dataset. (a) Ground-truth map. (b) SPRN. (c) SSFTT. (d) NSCKL. (e) GSCVit. (f) CLMG. (g) MS2FN. (h) SSEFN. (i) CSCAN. (j) PGC-SCTN.

TABLE XII
CLASSIFICATION RESULTS(%) ON LN01 DATASET WITH 20 LABELED SAMPLES PER CLASS

Class	SPRN(2022)	SSFTT(2022)	NSCKL(2023)	GSCVit(2024)	CLMG(2024)	MS2FN(2025)	SSEFN(2025)	CSCAN(2025)	PGC-SCTN
1	97.92±1.40	97.99±2.03	98.12±1.05	98.22±1.31	98.84±0.95	98.73±1.08	99.33±1.04	**99.45±0.55**	98.95±1.11
2	90.66±3.00	88.24±1.91	**90.88±2.83**	53.59±43.49	88.50±3.20	89.38±3.53	89.15±2.42	89.34±1.70	89.20±0.87
3	92.36±7.44	90.38±6.59	89.12±6.79	88.83±11.39	91.25±9.06	92.61±2.64	**97.64±2.11**	93.84±3.05	93.53±1.35
4	98.89±1.63	99.67±0.47	92.98±4.41	98.75±0.91	73.11±27.25	95.84±3.26	98.64±0.97	99.64±0.12	**99.73±0.20**
5	83.87±5.12	84.70±4.91	77.40±6.49	85.65±4.34	**86.38±4.28**	86.04±8.25	69.12±30.07	84.42±3.60	**86.38±1.66**
6	**91.78±3.70**	86.60±3.92	83.91±3.48	87.13±3.71	83.26±11.02	89.40±4.14	54.95±23.66	89.39±5.40	89.21±3.20
7	84.20±15.49	89.80±8.12	75.51±6.17	87.24±8.08	72.91±26.04	91.59±1.62	59.60±29.63	91.44±6.30	**94.75±2.82**
8	83.53±7.26	92.67±3.46	75.35±9.06	94.29±2.48	88.02±7.16	92.18±5.65	59.23±22.13	**96.54±1.64**	93.06±2.95
9	95.99±2.68	97.52±1.56	89.80±3.15	94.69±2.78	94.50±3.42	95.68±3.84	**98.84±0.86**	97.31±1.32	97.75±1.35
10	88.23±7.10	87.63±3.29	85.74±3.63	92.10±3.65	89.86±1.53	90.81±9.53	84.27±10.36	90.91±2.92	**92.32±1.34**
OA	92.05±0.97	91.73±0.57	88.74±1.55	77.96±17.48	90.75±1.84	92.00±1.52	90.14±1.76	92.67±0.60	**92.90±0.14**
AA	90.74±2.40	91.52±1.01	85.88±0.67	88.09±5.49	86.66±6.63	92.23±1.09	81.08±3.47	93.26±0.87	**93.49±0.71**
kappa	89.10±1.27	88.68±0.77	84.68±1.98	73.20±19.81	87.40±2.41	89.06±2.00	86.43±2.41	89.96±0.79	**90.28±0.19**

presented in Table XII, and the classification maps of different methods are illustrated in Fig. 11. Experimental results demonstrate that the proposed PGC-SCTN achieves the highest OA, AA, and Kappa coefficients on the challenging LN01 dataset. This further validates that the model can effectively capture discriminative spectral–spatial features and maintain superior generalization ability in complex real-world remote sensing scenarios.

IV. CONCLUSION

In this article, we propose a PGC-SCTN to address the challenge of limited labeled samples in HSI classification. The proposed framework integrates three core modules to achieve progressive optimization of spectral–spatial features. The DCE module excavates fine-grained local features through stacked convolutions. The PGFE module adopts horizontal–vertical separable convolution to efficiently model long-range dependencies. The SCT module decomposes 2-D global attention into 1-D chunkwise computations, reducing the complexity from $O((HW)^2)$ to $O(HW*(H + W))$ and balancing global semantic refinement with computational efficiency. Extensive experiments on four public HSI datasets (Indian Pines, Salinas Scene, Houston 2013, and Fanglu Tea Farm) demonstrate the superiority of PGC-SCTN over other classical models. Ablation studies further validate the effectiveness of different components.

Although the PGC-SCTN algorithm proposed in this study exhibits excellent performance in few-shot HSI classification, it has not yet fully exploited the potential value of the large number of unlabeled samples in hyperspectral data. In future research, we will deeply integrate semisupervised and weakly supervised learning theories to construct a sample utilization mechanism suitable for hyperspectral scenarios. By effectively activating the information value of unlabeled samples, we will further improve classification accuracy and robustness under the condition of limited labeled samples. We will also explore self-supervised pretraining strategies tailored to the characteristics of hyperspectral data, optimize the initialization and representation capability of the feature extraction network, and enhance the model's generalization performance in scenarios with extremely few labels.

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